Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines offer a scalable approach to quantum generative modeling by decoupling model optimization from quantum hardware. However, they are currently restricted to modeling single distributions. In this work, we introduce a conditional formulation for IQP Born machines that retains their classical trainability. By encoding the conditioning variable into a state that supports efficient classical sampling and amplitude-ratio estimation, we enable the scalable computation of Pauli-Z moments and their gradients. We empirically validate this approach on a sixteen-qubit discrete Gaussian family and the Curie-Weiss model. We also demonstrate the computational feasibility of classical conditional training on 64 to 2048 qubits, with residual errors in held-out sampled maximum mean discrepancy (MMD). Finally, we show how this encoding mechanism can be repurposed to induce symmetries in the output distributions of unconditional Born machines. This formulation extends classically trainable IQP Born machines to conditional generation.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ with a model $q_{\boldsymbol{\theta}}(\cdot \mid c)$ whose learned parameters are shared across conditions. Classical examples include conditional generative adversarial networks [1–3], conditional variational autoencoders [4–6] and conditional diffusion models [7–9]. In the quantum domain, quantum circuit Born machines (QCBMs) [10, 11] have also been adapted to conditional tasks [12]. The main strategy encodes the conditioning variables into the initial quantum state before a shared parameterized circuit [13, 14]. Related hybrid frameworks encode labels on additional qubits [15] or use a condition register to select time-dependent output distributions [16]. An alternative learns a joint Born distribution and obtains conditional samples by postselecting outcomes consistent with observed variables [17]. Beyond QCBMs, quantum discrete diffusion models support conditional image completion by fixing observed bits during denoising [18].

A particularly promising architecture in this space is the instantaneous quantum polynomial-time (IQP) Born machine. Sampling from certain IQP output families is conjectured to be classically intractable under standard complexity-theoretic assumptions [19–23]. Yet, for training via the maximum mean discrepancy (MMD) objective [24], which is common for QCBMs [10, 25], the required Pauli-Z expectation values and parameter gradients admit efficient classical estimators [26]. These estimators build on Van den Nest's techniques for computationally tractable states [27]. However, the standard formulation models a single target distribution per circuit. Extending IQP Born machines to conditional generation requires a conditioning mechanism that preserves classical access to the training observables.

In this work, we formulate a conditional IQP Born machine in which a condition-dependent input state $|\phi(c)\rangle$ is followed by one shared learned unitary $U_{\boldsymbol{\theta}}$, giving $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. Prior works have already explored conditional input preparation for quantum models [13, 14]. Our contribution is to identify and construct encoders whose reference amplitudes $\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$ permit efficient sampling and amplitude-ratio evaluation across the queried conditions. Applying existing sparse-observable methods [27, 28] and IQP training methods [26], we obtain unbiased estimators of each conditional Pauli-Z moment and phase derivative, with second moments bounded by one and four, respectively. Each specified quantity therefore admits additive error ϵ at fixed confidence using $O(\epsilon^{-2})$ reference samples. Figure 1 summarizes the construction and training objective.

We evaluate this conditional formulation across several tasks. First, a sixteen-qubit discrete Gaussian family illustrates how the shared model converts the input condition into a translated output profile. Second, the Curie–Weiss model tests temperature-dependent changes in distributional shape. Comparisons with independently fitted circuits reveal a tradeoff: independent fits have lower mean profile errors at $\beta = 0.40$ and 1.00, whereas the shared model has lower mean error at $\beta = 1.60$. We also assess the computational feasibility of classical training on up to 2048 qubits. Finally, we show how input-state encoding can impose strict output symmetries in uncon-

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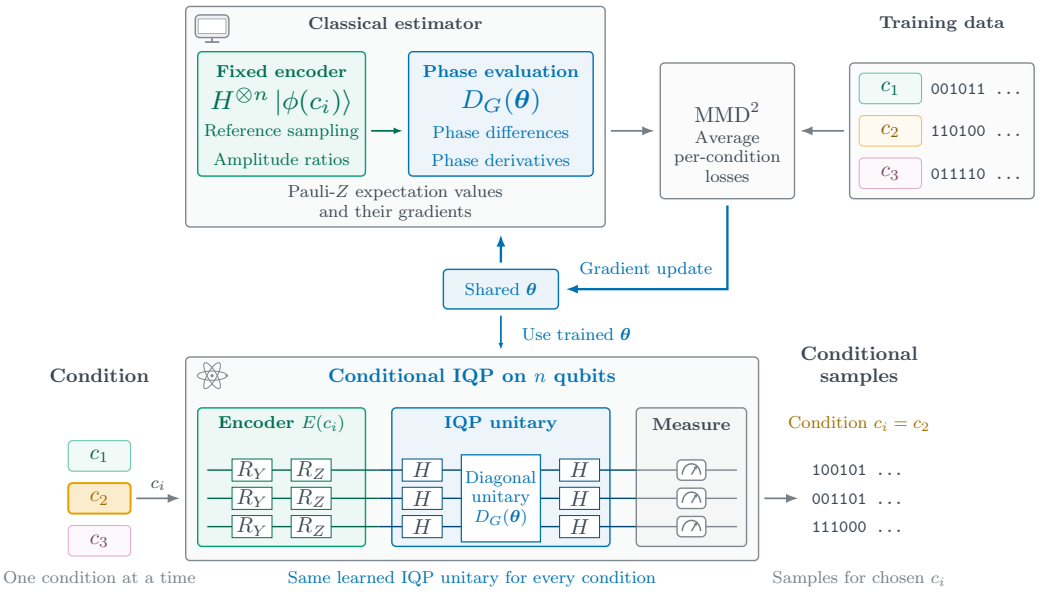


FIG. 1. Proposed framework for classical training of instantaneous quantum polynomial-time (IQP) Born machines. For each condition c_i, an encoder prepares a classically tractable input state $|\phi(c_i)\rangle$. The same IQP unitary $U_\theta = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}$, defined by a fixed graph G and a parameter vector $\boldsymbol{\theta}$, is used for all conditions. The conditional family is therefore represented by varying only the input state. During training, the per-condition discrepancies $\mathrm{MMD}^2(p(\cdot \mid c), q_\theta(\cdot \mid c))$ are averaged into a shared objective $\mathcal{L}(\theta) = |C|^{-1} \sum_{c \in C} \mathrm{MMD}_c^2$.

ditional IQP Born machines.

The rest of the paper is organized as follows: Section II presents the formulation of the conditional IQP model. Section III presents the conditional training protocol with the MMD. Section IV presents the experimental setup and the results from the numerical experiments. Finally, Sections VI and VII discuss the broader implications of our findings and conclude the work.

II. CONDITIONAL IQP MODEL

The proposed model separates the dependence on the condition from the learned unitary transformation. An encoder maps each $c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, while one parameterized IQP unitary $U_{\boldsymbol{\theta}}$ acts at every condition. For a target family $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings, the model distribution is

$$q_{\boldsymbol{\theta}}(x \mid c) = \left| \langle x | U_{\boldsymbol{\theta}} |\phi(c)\rangle \right|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

The encoder determines how the generative model responds to c; the parameterized ansatz determines the transformation shared by the family. Throughout this work, "IQP unitary" refers only to the learned block $U_{\boldsymbol{\theta}}$. The state-preparation circuit need not have IQP form, so the complete conditional circuit need not belong to the strict IQP circuit class. The central question is which condition-dependent input states preserve classical access to the observables and gradients required for training. The choice of encoder also restricts the conditional functions the model can represent. For rotation-based data encoding, the encoding generators determine accessible Fourier frequencies, while the learned circuit controls their coefficients [29]. Multidimensional Fourier analyses further characterize the dependence on encoding and circuit architecture [30]. These expressivity results motivate treating encoder design separately from the classical estimation criterion.

We parameterize the shared IQP unitary by an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, and each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. The unitary is

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j \right). \tag{3}$$

Because all Pauli-Z generators commute, the corresponding phase gates constitute a commuting diagonal layer. While the concrete implementation evaluated in this work employs at most two-body generators, this architecture naturally generalizes to arbitrary k-body interactions.

Classical training does not require samples from the circuit's output distribution. It instead uses the reference

state immediately before the diagonal IQP layer. Define

$$\psi_c(z) = \langle z| H^{\otimes n} |\phi(c)\rangle , \tag{4}$$

so that $|\psi_c(z)|^2$ is the computational-basis distribution of the reference state associated with condition c. We restrict attention to encoders for which, over the relevant conditions c and bit masks a, one can efficiently

1. sample $z \sim |\psi_c(z)|^2$, and
2. evaluate the amplitude ratio

$$R_c(z,a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{5}$$

for sampled z.

We refer to these two operations as the sampling-and-ratio interface.

These operations expose the two quantities changed by a Pauli-X mask. Sampling selects a reference-basis string z, while the ratio compares its amplitude with that of the paired string $z \oplus a$. The diagonal IQP layer contributes the phase difference between the same two strings. Their product gives one Monte Carlo contribution to a Pauli-Z moment [27].

The ratio is required only on the support of $|\psi_c|^2$, since strings with $\psi_c(z) = 0$ are never sampled. Computationally tractable states in the sense of Van den Nest [27], which admit efficient computational-basis sampling and amplitude evaluation, provide a sufficient realization of this interface under suitable numerical precision assumptions [27].

For a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^n |\phi_i(c)\rangle$, the reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b| H |\phi_i(c)\rangle$. Reference samples can therefore be generated by drawing each bit independently, and the amplitude ratio is

$$R_c(z,a) = \prod_{i:a_i=1} \frac{\psi_{c,i}(1-z_i)}{\psi_{c,i}(z_i)}. \tag{6}$$

When the local amplitudes have constant arithmetic evaluation cost, both operations require at most $O(n)$ local operations.

Efficient classical estimation of the observables used for training does not imply efficient classical sampling from $q_{\boldsymbol{\theta}}(\cdot \mid c)$. Existing IQP sampling-hardness results apply to specified circuit families and do not establish hardness for arbitrary condition-dependent input states or particular trained parameters. Furthermore, hardware sampling on a quantum computer requires efficient preparation circuits for both $|\phi(c)\rangle$ and the IQP unitary $U_{\boldsymbol{\theta}}$, which the classical sampling-and-ratio interface does not inherently provide.

A. Conditional Pauli estimator and gradients

The conditional MMD objective in Sec. III reduces to Pauli-Z moments. For a mask a, write $Z_a = \bigotimes_{i=1}^n Z_i^{a_i}$ and denote its model expectation by $\mu_q(a \mid c)$. We now show how reference-state access yields an estimator of this quantity and its phase derivatives.

To include ancillary qubits, let n be the number of visible qubits, r the number of ancillas, and $N = n + r$, allowing $r = 0$. An encoder prepares a normalized state $|\Phi(c)\rangle$ on the joint register, independently of the learned phase parameters. Only the bits corresponding to visible qubits are retained. We write the diagonal IQP layer as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle , \tag{7}$$

with

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P_N} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{8}$$

where $g_j \in \{0,1\}^N$ specifies the support of the jth diagonal Pauli generator.

For a visible mask $a \in \{0,1\}^n$, define its extension to the joint register as

$$\widetilde{a} = (a, 0^r), \tag{9}$$

and let

$$\psi_c(z) = \langle z| H^{\otimes N} |\Phi(c)\rangle \tag{10}$$

denote the corresponding reference amplitudes. The phase change induced by flipping the bits selected by $\widetilde{a}$ is

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus \widetilde{a}) - f_{\boldsymbol{\theta}}(z). \tag{11}$$

Using the amplitude ratio introduced in Eq. (5), define

$$\begin{aligned} W_{c,a}(z;\boldsymbol{\theta}) &= R_c(z,\widetilde{a}) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z;\boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}). \end{aligned} \tag{12}$$

Proposition 1 (Classical estimation from reference-state access)**.** *Fix a condition c, a visible mask a, and a polynomial-size diagonal Pauli phase function as in Eq. (8). Suppose that the normalized reference state is independent of $\boldsymbol{\theta}$, and that independent sampling from $|\psi_c|^2$ and evaluation of $R_c(z,\widetilde{a})$ on its sampled support are efficient. With exact sampling and arithmetic, the random variables*

$$\begin{aligned} Y &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}), \\ Y'_j &= \mathrm{Re}[i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta})] , \end{aligned} \tag{13}$$

where $d_{j,a}(z) = \chi_{g_j}(z \oplus \widetilde{a}) - \chi_{g_j}(z)$, satisfy

$$\begin{aligned} \mathbb{E}[Y] &= \mu_q(a \mid c), & \mathbb{E}[Y^2] &\le 1, \\ \mathbb{E}[Y'_j] &= \partial_{\theta_j} \mu_q(a \mid c), & \mathbb{E}[(Y'_j)^2] &\le 4. \end{aligned} \tag{14}$$

The means of S independent samples are therefore unbiased, with mean-square errors at most $1/S$ and $4/S$, respectively. Each specified moment or phase derivative can be estimated to additive error ϵ with fixed confidence using $S = O(\epsilon^{-2})$ samples. Full support is not required.

For computationally tractable reference states, the existence of efficient additive estimates also follows from Van den Nest's sparse-operator theorem [27]: $D_{\boldsymbol{\theta}}^{\dagger} X_{\widetilde{a}} D_{\boldsymbol{\theta}}$ and its phase derivatives have at most one nonzero entry per row and column, with norms at most one and two, respectively. Proposition 1 gives the explicit ratio estimator used here.

Proof. Conjugating the visible observable $Z_a \otimes I^{\otimes r}$ through the final Hadamard layer gives $X_{\widetilde{a}}$. The corresponding conditional moment is therefore

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c | \, D_{\boldsymbol{\theta}}^{\dagger} X_{\widetilde{a}} D_{\boldsymbol{\theta}} \, | \psi_c \rangle \\ &= \sum_z \psi_c(z)^* \psi_c(z \oplus \widetilde{a}) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)} \\ &= \mathbb{E}_{z \sim |\psi_c|^2} \left[Y_{c,a}(z; \boldsymbol{\theta}) \right]. \end{aligned} \tag{15}$$

The final equality follows from $R_c(z, \widetilde{a}) = \psi_c(z \oplus \widetilde{a}) / \psi_c(z)$ on the support of the reference distribution. Since the Pauli observable is Hermitian, its expectation value is real, so taking the real part does not change the result.

For the standard IQP input, the reference amplitudes are uniform and $R_c(z, \widetilde{a}) = 1$. Equation (15) then reduces to the usual cosine estimator [27, 31]. The amplitude ratio is the additional factor that accommodates the condition-dependent reference state.

In practice, we estimate Eq. (15) using Monte Carlo samples $z_s \sim |\psi_c|^2$,

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s; \boldsymbol{\theta}). \tag{16}$$

The amplitude ratio $R_c(z, \widetilde{a})$ can in principle become large when $\psi_c(z)$ is small. Nevertheless, the corresponding Monte Carlo estimator has controlled variance. Indeed, on the support $S_c = \{z : \psi_c(z) \neq 0\}$,

$$\begin{aligned} \mathbb{E}_{z \sim |\psi_c|^2} \left[|W_{c,a}(z; \boldsymbol{\theta})|^2 \right] &= \sum_{z \in S_c} |\psi_c(z)|^2 \left| \frac{\psi_c(z \oplus \widetilde{a})}{\psi_c(z)} \right|^2 \\ &= \sum_{z \in S_c} |\psi_c(z \oplus \widetilde{a})|^2 \leq 1. \end{aligned} \tag{17}$$

Thus, although individual ratio-weighted samples need not be bounded, their second moment is. Since $Y_{c,a} = \operatorname{Re} W_{c,a}$,

$$\operatorname{Var}(Y_{c,a}) \leq 1, \tag{18}$$

and the variance of the sample mean in Eq. (16) decreases as $1/S$. The conditional estimator therefore retains the usual $1/\sqrt{S}$ Monte Carlo convergence.

The gradients admit the same treatment. Because the encoder is fixed, neither the reference distribution nor the amplitude ratios depend on $\boldsymbol{\theta}$. Differentiating only the IQP phase gives

$$\begin{aligned} d_{j,a}(z) &= \chi_{g_j}(z \oplus \widetilde{a}) - \chi_{g_j}(z), \\ \partial_{\theta_j} \mu_q(a \mid c) &= \mathbb{E}_{z \sim |\psi_c|^2} \left[\operatorname{Re} \left(i \, d_{j,a}(z) W_{c,a}(z; \boldsymbol{\theta}) \right) \right]. \end{aligned} \tag{19}$$

Since $d_{j,a}(z) \in \{-2, 0, 2\}$, the corresponding gradient sample has second moment at most 4. Consequently, each phase derivative can also be estimated with $1/\sqrt{S}$ Monte Carlo convergence using the same reference samples as the moments.

These bounds do not require a full-support assumption: zero denominators are never sampled, and zero numerators evaluate to zero. The amplitude-ratio correction needed for condition-dependent inputs therefore does not introduce a divergent sampling variance.

Chebyshev's inequality gives failure probabilities at most $1/(S\epsilon^2)$ and $4/(S\epsilon^2)$ for the two sample means, proving the fixed-confidence statement. □

III. CONDITIONAL IQP TRAINING

This section formalizes the training of conditional IQP Born machines. Throughout this work, classically trainable means the training objective and its gradients can be estimated classically to additive accuracy.

A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population MMD is [24],

$$\begin{aligned} \mathrm{MMD}^2(p, q) = \mathbb{E}_{x,x' \sim p}[k(x, x')] - 2\mathbb{E}_{x \sim p,\, y \sim q}[k(x, y)] \\ + \mathbb{E}_{y,y' \sim q}[k(y, y')], \end{aligned} \tag{20}$$

where paired samples are drawn independently.

Throughout this work, we use the Gaussian (Hamming) kernel

$$k_\sigma(x, y) = \exp\left[-\frac{d_H(x, y)}{2\sigma^2} \right], \tag{21}$$

where d_H is the Hamming distance and $\sigma > 0$ is the kernel bandwidth.

Let $\mathcal{C}_{\mathrm{train}}$ denote the set of training conditions and τ the set of kernel bandwidths. We optimize the model with the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\mathrm{train}}|} \sum_{c \in \mathcal{C}_{\mathrm{train}}} \frac{1}{|\tau|} \sum_{\sigma \in \tau} \mathrm{MMD}_\sigma^2(p(\cdot \mid c), q_{\boldsymbol{\theta}}(\cdot \mid c)). \tag{22}$$

The objective weights all training conditions and bandwidths equally. At each optimizer update, we independently sample one condition and one bandwidth uniformly, giving a stochastic estimate of this average.

B. Character expansion

For binary strings, the Gaussian kernel admits an exact expansion in binary characters. Let $a \in \{0,1\}^n$ be a random visible-bit mask with independent coordinates

$$a_i \sim \text{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}, \tag{23}$$

and define the binary character

$$\chi_a(x) = (-1)^{a \cdot x}. \tag{24}$$

The kernel can then be written as

$$k_\sigma(x, y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]\,. \tag{25}$$

For a condition c, define the target and model character moments

$$\begin{aligned} \mu_p(a \mid c) &= \mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)], \\ \mu_q(a \mid c) &= \mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)]. \end{aligned} \tag{26}$$

Substituting Eq. (25) into Eq. (20) gives

$$\text{MMD}^2_\sigma(p(\cdot \mid c), q_{\boldsymbol{\theta}}(\cdot \mid c)) = \mathbb{E}_a\left[\left(\mu_p(a \mid c) - \mu_q(a \mid c)\right)^2\right]. \tag{27}$$

For a Born machine, $\chi_a(x)$ is the eigenvalue of the visible Pauli operator

$$Z_a = \bigotimes_{i=1}^{n} Z_i^{a_i}, \tag{28}$$

and therefore

$$\mu_q(a \mid c) = \langle Z_a \rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{29}$$

The conditional MMD can thus be expressed entirely in terms of visible Pauli-Z moments. This representation allows classical training to leverage the IQP expectation-value estimator detailed in Sec. II A.

The bandwidth σ determines which correlation orders are weighted most strongly. Since the mask coordinates are sampled independently, its Hamming weight $K = |a|$ follows

$$\begin{aligned} K &\sim \text{Binomial}(n, p_\sigma), \\ \mathbb{E}[K] &= np_\sigma, \\ \text{Var}(K) &= np_\sigma(1 - p_\sigma). \end{aligned} \tag{30}$$

Thus, each bandwidth induces a distribution over Pauli-string weights rather than selecting a single correlation order. Large bandwidths favor low-weight observables, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The choice of bandwidth has large implications, as it dictates not only which data correlations are weighted by the loss, but also the overall trainability of the model. Previous results demonstrate that cost functions consisting of global terms can induce barren plateaus [32, 33], even in shallow quantum circuits [34]. This creates a fundamental trade-off. Ensuring trainability may require restricting the objective to low-body correlations, which in turn limits the objective's ability to distinguish distributions that differ in higher-order correlations [35]. While various strategies have been proposed to navigate these optimization issues—including tailored generator sets, restricted initializations [36, 37], and adversarial kernel adaptation [38]—this remains an open question outside the scope of this work.

C. Unbiased finite-sample MMD and gradients

For a fixed condition c and visible mask a, let

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c), \tag{31}$$

for $B \geq 2$ independent target samples, and let

$$Y_s = Y_{c,a}(z_s; \boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2, \tag{32}$$

for $S \geq 2$ independent reference samples drawn independently of the target samples, with $Y_{c,a}$ defined in Eq. (12). Define the corresponding sample means

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{33}$$

A direct estimate of the squared moment difference, $(\overline{Y} - \overline{T})^2$, is biased because squaring a finite-sample mean introduces its sampling variance. We remove this bias by forming products only between independent sample pairs,

$$\begin{aligned} \widehat{\Delta^2_c}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}. \end{aligned} \tag{34}$$

Independence of the paired samples gives

$$\mathbb{E}\left[\widehat{\Delta^2_c}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{35}$$

Thus, Eq. (34) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (27).

We do not need to evaluate the pair sums explicitly. They can be written as

$$\sum_{s\neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{36}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{37}$$

We can therefore evaluate the estimator in linear rather than quadratic time in the numbers of target and reference samples.

The same construction gives an unbiased estimator of the phase gradients. Let

$$Y'_s = \partial_{\theta_j} Y_{c,a}(z_s;\boldsymbol{\theta}), \tag{38}$$

with Y'_s evaluated using Eq. (19). Differentiating Eq. (34) gives

$$\partial_{\theta_j}\widehat{\Delta^2_c}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y'_t - 2\overline{T}\,\overline{Y'}, \tag{39}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_s Y'_s. \tag{40}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta^2_c}(a)\right] = 2\left(\mu_q(a\mid c) - \mu_p(a\mid c)\right)\partial_{\theta_j}\mu_q(a\mid c), \tag{41}$$

which is the gradient of the corresponding squared moment difference. At a fixed parameter vector, these population unbiasedness statements average over independent target and reference samples. During training, target minibatches are sampled with replacement from fixed training pools; the stochastic gradient is therefore unbiased for the corresponding empirical objective. An unbiased finite-sample estimate of MMD^2 may be negative.

Excluding the diagonal terms $s = t$ is important because evaluating Y_s and Y'_s from the same reference draw correlates them. Since the encoder is fixed, the reference sampling law does not depend on $\boldsymbol{\theta}$; consequently, holding the sampled reference strings fixed during automatic differentiation evaluates the required phase gradients without differentiating through the sampler.

D. Stochastic mask estimator

Equation (27) expresses the MMD as an expectation over binary masks a. Since the number of possible masks grows exponentially with n, we estimate this expectation by sampling a finite mask bank from the distribution in Eq. (23).

The identity mask $a = 0$ satisfies

$$\mu_p(0\mid c) = \mu_q(0\mid c) = 1 \tag{42}$$

for every condition and therefore contributes identically zero to the MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{43}$$

The total probability assigned to nonidentity masks under the original kernel distribution is

$$Q^+_\sigma = 1 - (1-p_\sigma)^n. \tag{44}$$

Multiplying the conditional expectation by Q^+_σ therefore preserves the original Hamming-RBF objective.

For M sampled masks $a_1,\ldots,a_M$, the finite-mask estimator at condition c is

$$\widehat{\mathrm{MMD}}^2_\sigma(c) = Q^+_\sigma \frac{1}{M}\sum_{m=1}^{M}\widehat{\Delta^2_c}(a_m), \tag{45}$$

where $\widehat{\Delta^2_c}$ is the unbiased finite-sample estimator in Eq. (34). Averaging also over the mask draws recovers the full kernel MMD.

We need not evaluate repeated masks more than once. If $\mathcal{U}$ denotes the set of distinct sampled masks and m_u the multiplicity of mask u, Eq. (45) can be written equivalently as

$$\widehat{\mathrm{MMD}}^2_\sigma(c) = Q^+_\sigma \sum_{u\in\mathcal{U}}\frac{m_u}{M}\widehat{\Delta^2_c}(u). \tag{46}$$

Combining duplicate masks, therefore, reduces the number of Pauli moments we must evaluate without changing either the sampled objective or its gradient.

IV. NUMERICAL EXPERIMENTS

We evaluate the proposed architecture across two benchmarks: a translated Gaussian family to illustrate the conditioning mechanism and the Curie-Weiss model to test changes in distributional shape. Additionally, we provide a supplementary 2048-qubit scaling demonstration. For the first two experiments, we use $n = 16$ visible qubits and report the exact total variation distance between Hamming-weight profiles,

$$\mathrm{TVD}_K(p,q) = \frac{1}{2}\sum_{k=0}^{n}|\rho_p(k) - \rho_q(k)|, \tag{47}$$

where $\rho(k)$ is the probability of observing a bitstring of weight k. The targets assign equal probability within each Hamming-weight shell. The learned circuit need not preserve this property, so profile TVD measures the weight distribution but does not resolve errors within shells.

A. Discrete Gaussian translation

Our first experiment illustrates the conditioning mechanism with a moving Gaussian profile. For $n = 16$ bits, let $K = |x|$ be the Hamming weight. The target distribution translates the nominal center of a discrete Gaussian based on a condition $c \in [0,1]$, with all bitstrings of equal weight having equal target probability.

We use a fixed product state encoder that applies the same two rotations to every qubit:

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z\left(c - \frac{1}{2}\right) R_Y(1.50)\,|0\rangle_i\,. \tag{48}$$

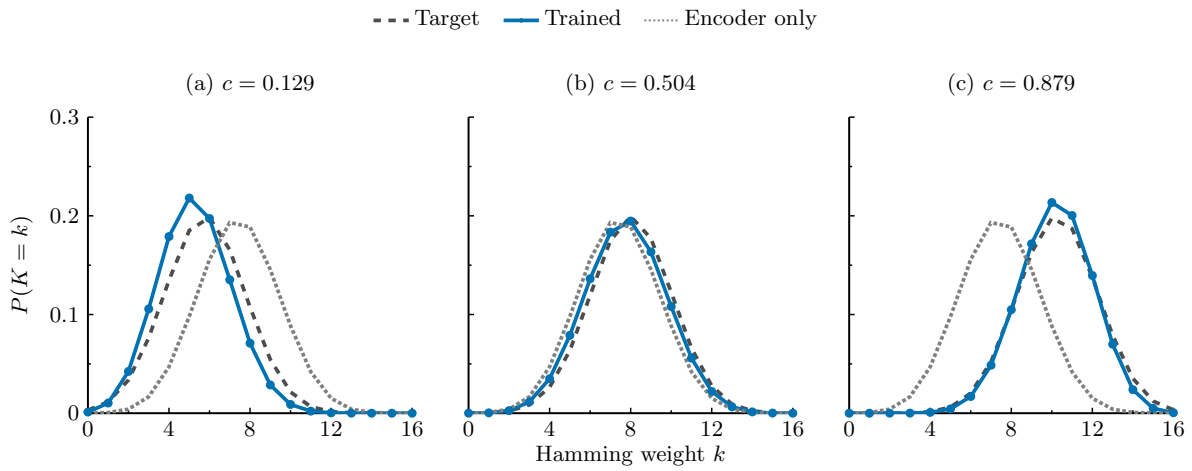


FIG. 2. Mean Hamming weight translation via conditioning. The three panels compare the target, the trained conditional IQP Born machine, and the encoder-only output. The displayed conditions were excluded from training and checkpoint selection. Dashed curves show the targets, solid blue curves show the trained model, and dotted grey curves show the encoder-only output. Model curves illustrate one run selected independently of held-out performance; the aggregate error reported in the text summarizes ten runs.

Measurement immediately after this encoder yields a fixed distribution; changing c modifies the input phases but leaves the outcome probabilities unchanged.

The shared unitary uses a 4×4 lattice with 16 one-body gates and 24 horizontal and vertical nearest-neighbor gates, giving 40 trainable parameters.

We train at $c \in \{0, 0.250, 0.500, 0.750, 1\}$ using 1000 target samples per condition and 300 optimizer updates. We evaluate the selected circuits across 80 conditions excluded from training. The median across runs of the mean evaluation profile TVD is 0.0670, compared with 0.292 for the encoder alone. Figure 2 illustrates the learned translation by the conditional model.

B. Curie-Weiss spin model

The Curie–Weiss model [39] tests the architecture's ability to capture changes in distributional shape. The spins $s_i \in \{-1, +1\}$ interact with equal strength, so the target assigns equal probability to bitstrings with the same Hamming weight. As the inverse temperature β increases, the finite-size Hamming-weight profile changes from a central peak to two symmetric modes.

We encode the inverse temperature using the fixed product phase encoder

$$|\phi(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.750\beta) R_Y\left(\frac{\pi}{2}\right) |0\rangle_i \,. \tag{49}$$

The encoder-only output is uniform for every β. The parameterized circuit uses all one- and two-body generators, giving 136 parameters. We train at $\beta \in \{0.60, 0.80, 1.20, 1.40\}$ and evaluate at $\beta \in \{0.40, 1.00, 1.60\}$.

We compare the shared conditional IQP across conditions with two independently tuned baseline families. Each family fits three separate circuits, one at each reporting temperature. The independent conditional IQP per condition retains the corresponding temperature encoder, while the independent unconditional IQP per condition uses the ordinary input.

We select shared-model hyperparameters and checkpoints using exact profile TVD on development temperatures $\beta \in \{0.50, 0.70, 1.30, 1.50\}$, without changing the fitting temperatures or the 10,000 training samples per temperature. Each independent family uses one Adam configuration selected across all three fitted temperatures. All final trained models use five fresh seeds with three restarts per circuit. Shared-model reporting temperatures are omitted from fitting and current selection, but were examined in earlier experiments; they are not a wholly untouched test. Appendix A 3 a specifies the HPO protocol.

Table I reports mean exact profile TVD and sample standard deviation across five seeds. The shared conditional IQP across conditions improves mean profile error over the encoder at all three reporting temperatures. The independent fits have lower mean errors at $\beta = 0.40$ and 1.00, while the shared conditional IQP across conditions has lower mean error at $\beta = 1.60$. At this highest inverse temperature, shared-model errors range from 0.0843 to 0.374, compared with 0.263–0.353 for independent conditional IQP per condition and 0.342–0.362 for independent unconditional IQP per condition. The shared conditional IQP across conditions has lower profile error than each independent family in four of the five paired seeds there. These comparisons describe this target family and protocol, rather than a general advantage of parameter

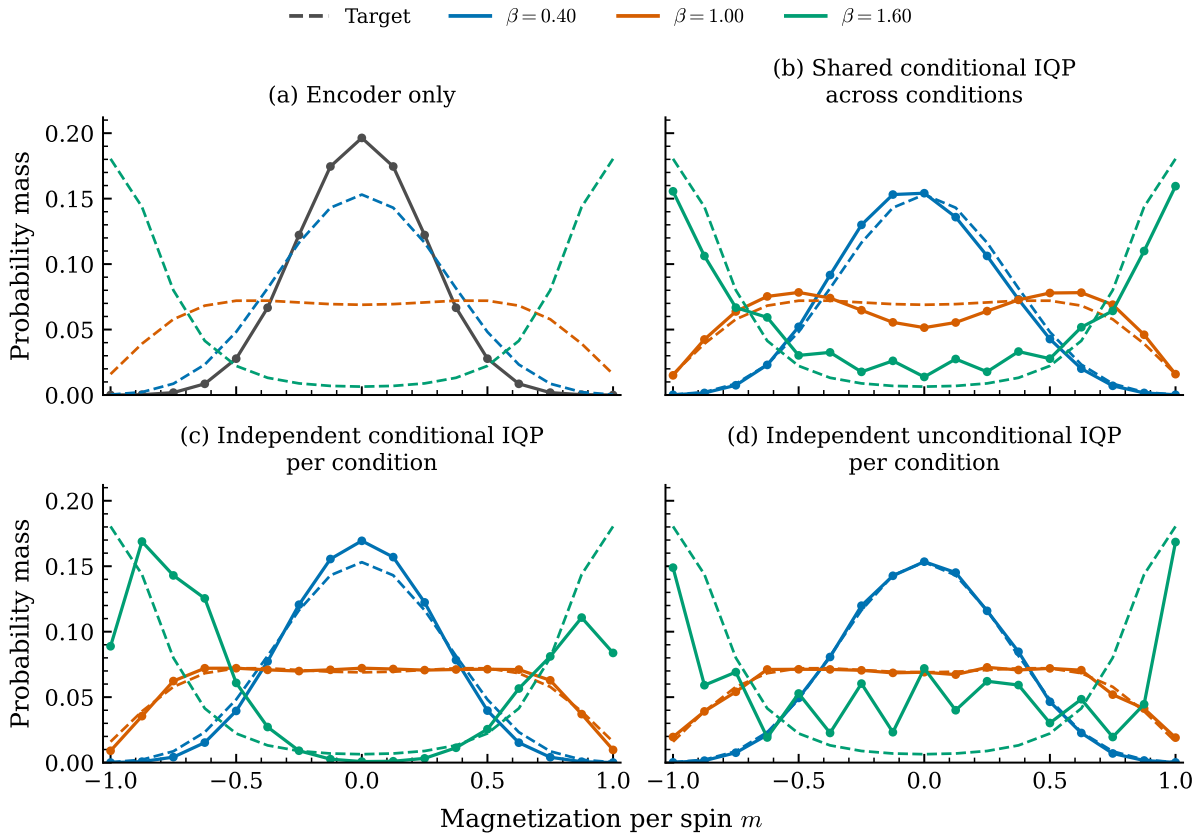


FIG. 3. Curie–Weiss magnetization profiles at sixteen qubits, with dimensionless magnetization per spin $m = (2k - 16)/16$. Panels show (a) encoder only, (b) shared conditional IQP across conditions, (c) independent conditional IQP per condition, and (d) independent unconditional IQP per condition. Solid curves show mean exact raw profiles over the same five independent seeds; the encoder is deterministic. Dashed curves show exact targets at $\beta = 0.40, 1.00, 1.60$. The shared conditional IQP across conditions excludes these temperatures from fitting and current hyperparameter and checkpoint selection; independent circuits fit and select checkpoints at their displayed temperature.

sharing.

The comparison has equal trial counts and per-circuit fitting budgets, but unequal aggregate compute and evaluation roles. Per final seed, the shared conditional IQP across conditions trains one circuit with 40,000 pool entries and 6,000 updates across restarts; each independent family trains three circuits with 30,000 pool entries and 18,000 updates. Independent checkpoint selection minimizes exact profile TVD at the fitted temperature, so its reported errors are in-sample.

To assess errors hidden by Hamming-shell aggregation, we also compute the exact full-bitstring TVD, $\mathrm{TVD}(p, q) = \frac{1}{2}\sum_x |p(x) - q(x)|$, from the same selected circuits. At $\beta = 0.40, 1.00, 1.60$, the shared model has mean full-bitstring TVDs of 0.236, 0.210, and 0.234, respectively. These values exceed the corresponding mean profile TVDs, since aggregating probabilities by Hamming weight can hide errors within shells. The target is uniform within each shell, but the learned circuit need not preserve this property. The full-bitstring diagnostics therefore show remaining distributional errors despite the improved profiles. Appendix A 3 a reports the corresponding variability and baseline values; these diagnostics do not enter model selection.

C. Scaling

The scaling experiment retains its original optimizer and evaluation settings, which differ from the tuned sixteen-qubit comparison. It assesses computational feasibility rather than transfer of the tuned configuration. We have not tested such transfer.

We evaluate the computational feasibility of classical training at $n \in \{64, 128, 256, 512, 1024, 2048\}$ visible qubits, without ancillas. We retain the fixed product phase encoder $R_Z(0.750\beta)R_Y(\pi/2)$ on every qubit. The parameterized unitary contains independent one-body phases and horizontal and vertical nearest-neighbor pair phases on an open rectangular lattice. For an $L_x \times L_y$ lattice, the ansatz has $3n - L_x - L_y$ parameters, so the parameter count grows linearly with the number of qubits.

TABLE I. Exact raw Curie–Weiss Hamming-profile TVD at sixteen qubits. Trained rows report mean ± sample standard deviation over five independent seeds, with three restarts per circuit. The encoder-only baseline is deterministic. Shared-model temperatures are excluded from fitting and current selection, but have historical exposure; independent circuits fit and select at their displayed temperature.

Model	Evaluation	Hamming-profile TVD ↓		
		$\beta = 0.40$	$\beta = 1.00$	$\beta = 1.60$
Encoder only	Baseline	0.118	0.441	0.870
Shared conditional IQP across conditions	Held out	0.0712 ± 0.0158	0.0683 ± 0.0153	0.153 ± 0.124
Independent conditional IQP per condition	Fitted	0.0530 ± 0.0169	0.0334 ± 0.00884	0.296 ± 0.0377
Independent unconditional IQP per condition	Fitted	0.0170 ± 0.0114	0.0296 ± 0.00795	0.350 ± 0.00936

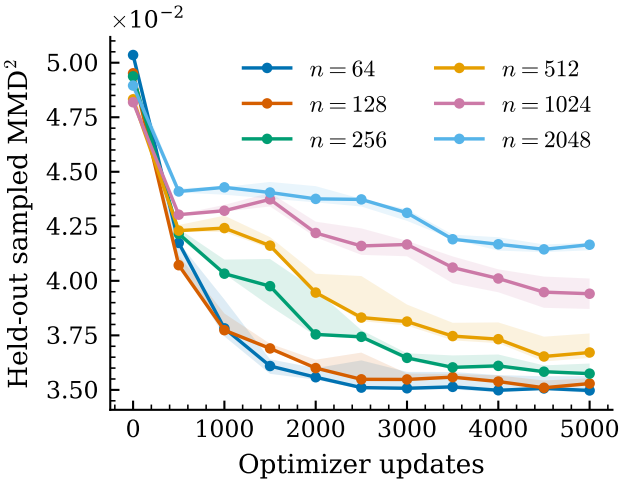


FIG. 4. Classical training of conditional IQP Born machines from 64 to 2048 qubits. Held-out sampled MMD^2 is evaluated at initialization and every 500 optimizer updates along the validation-selected restart for each run. Lines and shaded bands show the median and interquartile range across three independent runs after averaging evaluation banks within each run. Evaluation conditions are $\beta \in \{0.40, 0.95, 1.00, 1.05, 1.60\}$. The same bodyness targets $w \in \{2, 4\}$ are used at every system size, implying different kernel bandwidths.

The lattice dimensions and optimization hyperparameters are listed in Appendix A.

At each size, we train at $\beta \in \{0.60, 0.80, 1.20, 1.40\}$ using independent training runs and restarts, with 5000 optimizer updates per restart. Checkpoint selection uses sampled validation MMD^2 averaged over the training temperatures and bandwidths. We independently evaluate the selected restart at initialization and every 500 updates on $\beta \in \{0.40, 0.95, 1.00, 1.05, 1.60\}$, which are excluded from fitting and checkpoint selection. At each reporting point, we average independent evaluation banks within each run and summarize the averages by their median and interquartile range.

Figure 4 shows a decrease in held-out sampled MMD^2 at every system size, followed by a persistent residual. At 64 qubits, the median across runs decreases from 0.0504 at initialization to 0.0350 after 5000 updates; at 2048 qubits, it decreases from 0.0490 to 0.0417. These results demonstrate the feasibility of classically optimizing conditional IQP Born machines up to 2048 qubits. However, substantial discrepancies remain at high inverse temperature, and a decrease in sampled MMD^2 does not establish accurate recovery of the full target distribution. The same bodyness targets are used at every size, so the corresponding kernel bandwidths vary with system size. Absolute MMD^2 values should therefore not be interpreted as direct comparisons of distributional fidelity across sizes.

V. INPUT-STATE INDUCED SYMMETRY IN UNCONDITIONAL IQP BORN MACHINES

Encoder design determines both the dependence on a condition and the constraints satisfied by the output law. The conditional experiments illustrate the first role; we now demonstrate the second by preparing an input that enforces an exact bit-flip symmetry for every learned phase vector while preserving classical trainability. Such constraints provide inductive biases in variational quantum models [40].

Choose any nonzero bit mask $s \in \{0,1\}^n$ and prepare

$$|\phi_s\rangle = \frac{|0^n\rangle + |s\rangle}{\sqrt{2}}. \tag{50}$$

This state satisfies $X_s |\phi_s\rangle = |\phi_s\rangle$, where $X_s = \bigotimes_{i=1}^n X_i^{s_i}$ flips the qubits selected by s. Since the IQP unitary commutes with X_s, its output remains invariant under the same flip, even when all learned phases vary independently. Consequently,

$$q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus s) \qquad \text{for every } x \text{ and } \boldsymbol{\theta}. \tag{51}$$

The reference amplitudes are uniform on the strings satisfying $s \cdot z = 0$ and vanish elsewhere, evaluating the dot product modulo two. Sampling uniformly subject to this single parity constraint is efficient. For a sampled

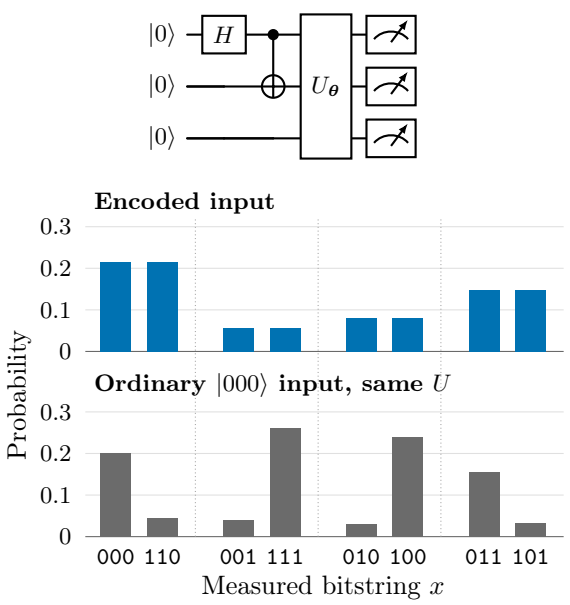


FIG. 5. The encoded input makes paired outcomes equally likely. The circuit prepares $(|000\rangle + |110\rangle)/\sqrt{2}$ before applying $U_{\boldsymbol{\theta}}$. The plots show exact probabilities at one parameter setting for the encoded input (blue) and the ordinary $|000\rangle$ input (gray), using the same unitary in both cases. Adjacent outcomes differ by 110. Their equality for the encoded input holds for every phase vector.

string z, the amplitude ratio is

$$R_s(z,a) = \begin{cases} 1, & s\cdot a = 0, \\ 0, & s\cdot a = 1. \end{cases} \tag{52}$$

The encoded input therefore satisfies the sampling-and-ratio interface, preserving classical access to the moments and gradients required for training.

For example, $s = 110$ on three qubits gives $|\phi_s\rangle = (|000\rangle + |110\rangle)/\sqrt{2}$ and enforces $q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus 110)$. The ordinary $|000\rangle$ input does not enforce this relation: with only $\theta_1 = \pi/6$ nonzero, it gives $q(000) = 3/4$ and $q(110) = 0$. Figure 5 compares the two inputs under the same unitary. This selected-subset construction illustrates the commuting-symmetry recipe of Recio-Armengol *et al.*, whose demonstrated example enforces global bit-flip symmetry [26].

VI. DISCUSSION

Our numerical experiments show that a shared parameter vector can learn condition-dependent output profiles. In the Curie–Weiss comparison, the shared conditional IQP across conditions improves mean profile TVD over the encoder at all three reporting temperatures. The independent conditional IQP per condition and independent unconditional IQP per condition have lower mean errors at $\beta = 0.40$ and 1.00, whereas the shared model has lower mean error at $\beta = 1.60$, with substantial variability across seeds. This outcome motivates investigating whether training across conditions can improve difficult fits. Different evaluation roles and unequal aggregate training budgets prevent attributing the observed gain to parameter sharing alone.

The full-bitstring diagnostics show that accurate Hamming-weight profiles can coexist with appreciable errors within shells. Efficient estimation of the training moments therefore does not guarantee accurate recovery of the full distribution. Herrero-Gonzalez *et al.* analyze discrepancies between classical correlator-based surrogates and deployed Born distributions [41]. Under the stated sampling and arithmetic assumptions, our estimators target the moments of the specified circuit without truncating Pauli propagation. Nevertheless, finite sampling budgets and the correlations emphasized by the kernel can leave distributional errors unresolved. Kernel-adaptive IQP training offers a possible extension beyond the fixed Gaussian kernels used here [38]; its benefit for conditional generation remains to be evaluated.

The scaling experiment demonstrates the computational feasibility of classical conditional training up to 2048 qubits, with residual sampled MMD^2 discrepancies. It uses different optimizer settings and a different ansatz from the tuned sixteen-qubit comparison, so transfer of the selected hyperparameters remains untested. The fixed bodyness targets imply size-dependent bandwidths, and the resulting MMD^2 values do not provide direct comparisons of distributional fidelity across sizes. Further study of the encoder, ansatz and optimization settings, together with diagnostics beyond sampled MMD, is needed to assess conditional accuracy at larger sizes.

Encoder design controls the conditional functions available to the model and can impose exact constraints on its output. The product phase encoders studied here provide simple state preparation and efficient sampling and amplitude-ratio evaluation. More structured encoders could offer richer conditional dependence if they retain this interface. The construction in Sec. V demonstrates a complementary role: input preparation restricts the output family to distributions with an exact bit-flip symmetry, independently of the learned phases. Spectral Born machines extend IQP generative models through group Fourier analysis while preserving efficient classical training [42]. Investigating conditional reference-state access in that setting could broaden the framework to other forms of discrete data.

Quantum deployment requires assessing how noise and hardware constraints affect conditional accuracy. Ryu *et al.* propose logical-noise injection to decompose large IQP circuits into smaller clusters that fit qubit-limited processors [43]. This approximate strategy introduces a further potential discrepancy from the trained distribution. Its suitability for conditional models requires evaluating such errors across conditions.

Whether useful trained conditional models retain classical sampling hardness remains open. Herbst *et al.* iden-

tify average-case trainability limitations for anticoncentrating output families [44]. Bläser *et al.* propose a cross-entropy verification protocol for cubic IQP circuits with planted independent spaces [45]. Under two complexity conjectures, secret knowledge of the planted structure creates a polynomial gap between verification and classical spoofing costs, although both remain exponential. These guarantees concern a prescribed circuit family and do not establish hardness for our trained conditional models. Practical quantum advantage would require evidence of useful conditional generation together with an appropriate classical comparison and a justified sampling-hardness claim.

VII. CONCLUSIONS

We introduced a conditional formulation for IQP Born machines that combines a condition-dependent encoder with a single shared learned unitary. Efficient reference-state sampling and amplitude-ratio evaluation enable unbiased classical estimation of conditional Pauli-Z moments and phase gradients, supporting optimization without quantum circuit evaluations. The same interface also accommodates input states that impose exact output symmetries.

The Gaussian and Curie–Weiss experiments demonstrate learned conditional profiles, with temperature-dependent tradeoffs against independently fitted circuits. Supplementary runs establish the computational feasibility of classical optimization up to 2048 qubits. Remaining full-bitstring TVD and sampled MMD^2 errors limit claims of distributional accuracy. More expressive tractable encoders, adaptive kernels and evaluation of quantum deployment are directions for improving conditional generation; classical sampling hardness and practical quantum advantage remain open.

DATA AVAILABILITY

The code and data supporting this study are available in GitHub [46].

ACKNOWLEDGMENTS

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[1] M. Mirza and S. Osindero, Conditional generative adversarial nets (2014), arXiv:1411.1784 [cs.LG].

[2] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, Image-to-image translation with conditional adversarial networks, in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2017) arXiv:1611.07004.

[3] T. Miyato and M. Koyama, cGANs with projection discriminator, in *International Conference on Learning Representations* (2018) arXiv:1802.05637.

[4] K. Sohn, H. Lee, and X. Yan, Learning structured output representation using deep conditional generative models, in *Advances in Neural Information Processing Systems*, Vol. 28 (2015).

[5] J. Walker, C. Doersch, A. Gupta, and M. Hebert, An uncertain future: Forecasting from static images using variational autoencoders, in *Computer Vision – ECCV 2016*, Lecture Notes in Computer Science, Vol. 9911 (2016) pp. 835–851, arXiv:1606.07873.

[6] X. Yan, J. Yang, K. Sohn, and H. Lee, Attribute2Image: Conditional image generation from visual attributes, in *Computer Vision – ECCV 2016*, Lecture Notes in Computer Science, Vol. 9908 (2016) pp. 776–791, arXiv:1512.00570.

[7] P. Dhariwal and A. Nichol, Diffusion models beat GANs on image synthesis, in *Advances in Neural Information Processing Systems*, Vol. 34 (2021) arXiv:2105.05233.

[8] J. Ho and T. Salimans, Classifier-free diffusion guidance (2022), arXiv:2207.12598 [cs.LG].

[9] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, High-resolution image synthesis with latent diffusion models, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2022) pp. 10684–10695, arXiv:2112.10752.

[10] J.-G. Liu and L. Wang, Differentiable learning of quantum circuit born machine, Physical Review A **98**, 062324 (2018).

[11] M. Benedetti, D. Garcia-Pintos, O. Perdomo, V. Leyton-Ortega, Y. Nam, and A. Perdomo-Ortiz, A generative modeling approach for benchmarking and training shallow quantum circuits, npj Quantum Information **5**, 45 (2019).

[12] Z. He, L. Li, S. Zheng, Z. Huang, and H. Situ, A conditional generative model based on quantum circuit and classical optimization, International Journal of Theoretical Physics **58**, 1138 (2019).

[13] M. Benedetti, B. Coyle, M. Fiorentini, M. Lubasch, and M. Rosenkranz, Variational inference with a quantum computer, Physical Review Applied **16**, 044057 (2021), arXiv:2103.06720 [quant-ph].

[14] O. Kiss, M. Grossi, E. Kajomovitz, and S. Vallecorsa, Conditional born machine for monte carlo event generation, Physical Review A **106**, 022612 (2022).

[15] Q.-W. Zeng, H.-Y. Ge, C. Gong, and N.-R. Zhou, Conditional quantum circuit born machine based on a hybrid quantum–classical framework, Physica A: Statistical Mechanics and its Applications **618**, 128693 (2023).

[16] S. Certo, A. Pham, N. Robles, and A. Vlasic, Conditional generative models for learning stochastic processes, Quantum Machine Intelligence **5**, 42 (2023), arXiv:2304.10382 [quant-ph].

[17] J. Zeng, Y. Wu, J.-G. Liu, L. Wang, and J. Hu, Learning and inference on generative adversarial quantum circuits, Physical Review A **99**, 052306 (2019), arXiv:1808.03425 [quant-ph].

[18] C. Chen, Q. Zhao, M. Zhou, D. Niyato, Z. He, and H. Situ, Overcoming dimensional factorization limits in discrete diffusion models through quantum joint distribution learning, npj Quantum Information **12**, 49 (2026).

[19] D. Shepherd and M. J. Bremner, Instantaneous Quantum Computation, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences **465**, 1413 (2009), arXiv:0809.0847 [quant-ph].

[20] M. J. Bremner, R. Jozsa, and D. J. Shepherd, Classical simulation of commuting quantum computations implies collapse of the polynomial hierarchy, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences **467**, 459 (2011).

[21] M. J. Bremner, A. Montanaro, and D. J. Shepherd, Average-Case Complexity Versus Approximate Simulation of Commuting Quantum Computations, Physical Review Letters **117**, 080501 (2016).

[22] M. J. Bremner, A. Montanaro, and D. J. Shepherd, Achieving quantum supremacy with sparse and noisy commuting quantum computations, Quantum **1**, 8 (2017).

[23] S. C. Marshall, S. Aaronson, and V. Dunjko, Improved separation between quantum and classical computers for sampling and functional tasks, in *40th Computational Complexity Conference (CCC 2025)*, Leibniz International Proceedings in Informatics (LIPIcs), Vol. 339 (Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025) pp. 5:1–5:14, arXiv:2410.20935 [quant-ph].

[24] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, A kernel two-sample test, Journal of Machine Learning Research **13**, 723 (2012).

[25] B. Coyle, D. Mills, V. Danos, and E. Kashefi, The Born supremacy: Quantum advantage and training of an Ising Born machine, npj Quantum Information **6**, 60 (2020).

[26] E. Recio-Armengol, S. Ahmed, and J. Bowles, Train on classical, deploy on quantum: Scaling generative quantum machine learning to a thousand qubits (2026), arXiv:2503.02934v2 [quant-ph].

[27] M. Van den Nest, Simulating quantum computers with probabilistic methods, Quantum Information and Computation **11**, 784 (2011), arXiv:0911.1624 [quant-ph].

[28] V. Havlíček, Amplitude ratios and neural network quantum states, Quantum **7**, 938 (2023), arXiv:2201.09128 [quant-ph].

[29] M. Schuld, R. Sweke, and J. J. Meyer, The effect of data encoding on the expressive power of variational quantum-machine-learning models, Physical Review A **103**, 032430 (2021).

[30] B. Casas and A. Cervera-Lierta, Multidimensional fourier series with quantum circuits, Physical Review A **107**, 062612 (2023).

[31] E. Armengol and J. Bowles, IQPopt: Fast optimization of instantaneous quantum polynomial circuits in JAX (2026), arXiv:2501.04776v2 [quant-ph].

[32] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, Barren plateaus in quantum neural network training landscapes, Nature Communications **9**, 4812 (2018).

[33] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, and M. Cerezo, Barren plateaus in variational quantum computing, Nature Reviews Physics **7**, 174 (2025).

[34] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, Cost function dependent barren plateaus in shallow parametrized quantum circuits, Nature Communications **12**, 1791 (2021).

[35] M. S. Rudolph, S. Lerch, S. Thanasilp, O. Kiss, O. Shaya, S. Vallecorsa, M. Grossi, and Z. Holmes, Trainability barriers and opportunities in quantum generative modeling, npj Quantum Information **10**, 116 (2024).

[36] K. Shen, S. Pielawa, V. Dunjko, and H. Wang, Characterizing trainability of instantaneous quantum polynomial circuit born machines (2026), arXiv:2602.11042v2 [quant-ph].

[37] S. Lerch, J. Bowles, R. Puig, E. Armengol, Z. Holmes, and S. Thanasilp, IQP born machines under data-dependent and agnostic initialization strategies (2026), arXiv:2603.14576 [quant-ph].

[38] A. Kurkin, K. Shen, S. Pielawa, H. Wang, and V. Dunjko, Universality and kernel-adaptive training for classically trained, quantum-deployed generative models (2025), arXiv:2510.08476 [quant-ph].

[39] R. S. Ellis, *Entropy, Large Deviations, and Statistical Mechanics*, Classics in Mathematics (Springer, Berlin, Heidelberg, 2006).

[40] J. J. Meyer, M. Mularski, E. Gil-Fuster, A. A. Mele, F. Arzani, A. Wilms, and J. Eisert, Exploiting symmetry in variational quantum machine learning, PRX Quantum **4**, 010328 (2023).

[41] M. Herrero-Gonzalez, B. Coyle, K. McDowall, R. Grassie, S. Beentjes, A. Khamseh, and E. Kashefi, The born ultimatum: Conditions for classical surrogation of quantum generative models with correlators (2025), arXiv:2511.01845 [quant-ph].

[42] A. Huang, W. Maxwell, V. Belis, E. Peters, J. Pye, S. Jahangiri, and J. Bowles, Spectral born machines: Classically trainable quantum generative models for discrete data (2026), arXiv:2607.06675 [quant-ph].

[43] J.-Y. Ryu, W. Song, K. Bae, W. Lee, and I. Sohn, Deployment of large IQP circuit born machines on qubit-limited hardware using noise injection (2026), arXiv:2609.35163 [quant-ph].

[44] S. Herbst, I. Brandić, and A. Pérez-Salinas, Limits of quantum generative models with classical sampling hardness (2025), arXiv:2512.24801 [quant-ph].

[45] M. Bläser, M. Gullans, D. Hangleiter, Y. Liu, and Y. Qiao, Verifiable quantum advantage based on polynomials with planted structures (2026), arXiv:2609.39918 [quant-ph].

[46] O. Balló-Gimbernat, Conditional generation with IQP circuits: Code and data (2026), gitHub repository.

[47] D. P. Kingma and J. Ba, Adam: A method for stochastic optimization, in *International Conference on Learning Representations* (2015) arXiv:1412.6980 [cs.LG].

[48] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, Optuna: A next-generation hyperparameter optimization framework (2019), arXiv:1907.10902 [cs.LG].

[49] A. Defazio, X. A. Yang, H. Mehta, K. Mishchenko, A. Khaled, and A. Cutkosky, The road less scheduled (2024), arXiv:2405.15682 [cs.LG].

Appendix A: Hyperparameters used for the numerical experiments

1. Common training settings

Reported trained models minimize the unbiased squared-MMD estimator using Adam [47], with second-moment coefficient 0.999 and numerical stabilizer 10^{-8}. The first-moment coefficient is 0.9 for the Gaussian and original scaling experiments and is tuned for Curie–Weiss. Learning rates follow a cosine decay, and parameters are initialized independently from $\mathcal{N}(0, s_\theta^2)$. Table II lists the experiment-specific budgets. Restarts within each model and seed share target pools and stochastic sampling schedules.

Each update uniformly samples one training condition and one bandwidth, then draws a target minibatch with replacement from its training pool. Nonidentity masks are refreshed at every update, retaining the nonidentity-probability correction. Mask budgets include repetitions. Reference samples are independent of the target minibatch and shared across masks.

TABLE II. Training and sampled-test budgets per circuit. Updates are per restart. Curie–Weiss learning rates, initialization widths and first-moment coefficients are selected by HPO; Gaussian and scaling settings retain their original values. Tuned values are rounded to three significant figures; reproduction configurations retain full precision. Test budgets refer to sampled MMD diagnostics; small-system TVDs use exact probabilities.

Setting	Gaussian	Shared conditional IQP across conditions	Independent conditional IQP per condition	Independent unconditional IQP per condition	Scaling
Updates per restart	300	2,000	2,000	2,000	5,000
Restarts per model and seed	1	3	3	3	3
Final evaluation seeds per model	10	5	5	5	3
Training-pool samples per condition	1,000	10,000	10,000	10,000	10,000
Test target draws per condition/bank	1,000	2,000	2,000	2,000	2,000
Test MMD banks per seed and condition grid	1	1	1	1	5
Target minibatch size	100	450	450	450	450
Mask draws per update	1,024	2,048	2,048	2,048	512
Reference draws per update	1,024	2,048	2,048	2,048	1,024
Expected original mask weights	$\{1\}$	$\{2, 4, 6\}$	$\{2, 4, 6\}$	$\{2, 4, 6\}$	$\{2, 4\}$
Initial learning rate	5×10^{-3}	0.0401	0.0400	0.0324	0.01
Final learning rate	5×10^{-4}	3.26×10^{-3}	7.83×10^{-4}	3.65×10^{-4}	10^{-3}
Initial phase standard deviation s_θ	0.05	0.430	0.349	0.0758	0.1
First-moment coefficient b_1	0.9	0.968	0.812	0.975	0.9

For expected original mask weight w, the bandwidth is

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where n counts visible qubits and the expectation precedes conditioning on a nonidentity mask. Bandwidths contribute equally to the objective. All encoders are fixed during training, and rotation angles are in radians. All reported model profiles use raw circuit probabilities. Encoder-only baselines receive no training. Sampled-test target draws are independent of the training pools; these diagnostics do not determine the exact profile TVDs.

2. Discrete Gaussian translation

The model has 16 visible qubits on an open 4×4 lattice, with 16 single-qubit and 24 pair phases. Its input is

$$|\phi_{\mathrm{G}}(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c - 1/2) R_Y(1.50) \, |0\rangle_i \, . \tag{A2}$$

The target profile is $\rho(k \mid c) \propto \exp[-(k - (5 + 6c))^2/8]$, with uniform probabilities within each Hamming-weight shell. Training uses $c \in \{0, 0.250, 0.500, 0.750, 1\}$ and 1,000 samples per condition, giving 5,000 training samples per seed. We select the checkpoint minimizing mean exact profile TVD at development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$, including initialization and checks every 30 updates. Evaluation uses $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$, with 1,000 target draws per condition for auxiliary test-MMD diagnostics. Profile TVDs use exact probabilities; the aggregate error is the median across ten independent seeds of the mean error over evaluation conditions.

3. Curie–Weiss comparison

The target law is $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. All trained comparison models have 16 visible qubits and the same complete bank of 16 single-qubit and 120 pair phases. Conditional models use

$$|\phi_{\mathrm{CW}}(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.750\beta) R_Y(\pi/2) \, |0\rangle_i \, . \tag{A3}$$

The shared conditional IQP across conditions trains at $\beta \in \{0.60, 0.80, 1.20, 1.40\}$ and is evaluated at $\beta \in \{0.40, 1.00, 1.60\}$.

Independent baselines fit a separate circuit at each reporting temperature, using either the corresponding conditional input or $|0\rangle^{\otimes 16}$. Each circuit has 10,000 training samples and uses 2,000 updates per restart with three restarts in the final evaluation. Each baseline family therefore requires three circuits per seed. Table IV distinguishes these aggregate budgets. Independent models are evaluated at fitted temperatures, whereas the shared conditional IQP across conditions is evaluated at temperatures omitted from fitting and selection. Auxiliary test-MMD diagnostics use 2,000 independently drawn targets per reported temperature. Profile TVDs use exact probabilities; final results report means and sample standard deviations across five independent seeds.

a. Hyperparameter optimization

We use Optuna's tree-structured Parzen estimator [48] to tune cosine-decay Adam and the Gaussian initialization width, while fixing the encoder, the 136-parameter ansatz, the training pools, kernel weights, and sampling budgets. The shared-model search uses 48 trials, each completing 2,000 sampled-MMD updates on each of two screening seeds, with one restart and no early pruning. The score averages the best exact development-profile TVD across the two seeds. Checkpoint selection includes initialization and every 25 updates at development temperatures $\{0.50, 0.70, 1.30, 1.50\}$. We also tried Schedule-Free Adam [49] with the same HPO trial budget, but it did not outperform cosine-decay Adam on the development-profile TVD.

Table III gives the Adam search ranges. Each study uses 12 startup trials and batches of two proposals, with sampler updates ordered independently of completion order. The second-moment coefficient, numerical stabilizer, and weight decay are fixed at 0.999, 10^{-8}, and zero. We confirm the two best distinct Adam configurations on three fresh seeds, with three restarts per seed, and select the lowest mean development-profile TVD. The selected settings in Table II are then fixed for evaluation on five further independent seeds, with three restarts per seed and development-based checkpoint selection. Reporting temperatures $\{0.40, 1.00, 1.60\}$ do not enter this search or selection, although they had been examined in earlier experiments.

For the independent conditional IQP per condition and independent unconditional IQP per condition families, we run one Adam HPO study per family, with 48 trials and the same search ranges and per-circuit budgets. Each trial trains separate circuits at $\beta = 0.40, 1.00, 1.60$ on both screening seeds and averages their best fitted-temperature profile TVDs into one score. One configuration is therefore tuned for each family, rather than separately for each temperature. The original Adam settings and the selected shared-model settings are reference trials within each baseline budget; the shared-model search includes its original settings as a reference trial. We confirm the two best configurations per family on three fresh seeds with three restarts per circuit, then evaluate each selected configuration

TABLE III. Adam HPO search ranges for the Curie–Weiss models. Learning rates, initialization widths, and decay ratios are sampled logarithmically; b_1 is sampled uniformly.

Setting	Search range
Initial learning rate	$[3 \times 10^{-4}, 5 \times 10^{-2}]$
Gaussian initialization width s_θ	$[0.01, 0.7]$
First-moment coefficient b_1	$[0.8, 0.98]$
Final/initial learning-rate ratio	$[0.01, 0.3]$

TABLE IV. Curie–Weiss comparison budgets per seed, summed over fitting conditions. Training samples count target-pool entries, rather than repeated minibatch draws. Optimizer updates include all three restarts of every trained circuit.

Model	Trained circuits	Training samples	Optimizer updates
Encoder only	0	0	0
Shared conditional IQP across conditions	1	40,000	6,000
Independent conditional IQP per condition	3	30,000	18,000
Independent unconditional IQP per condition	3	30,000	18,000

on five further independent seeds with three restarts per circuit. Selection includes initialization and every 25 updates at each fitted temperature. Equal trial counts require three times as many circuit fits per independent family as for the shared-model Adam search. These evaluations remain in-sample with respect to temperature; fresh seeds measure optimization variability.

Full-bitstring TVD is a secondary exact diagnostic. At $\beta = 0.40, 1.00, 1.60$, the shared conditional IQP across conditions gives 0.236 ± 0.0293, 0.210 ± 0.0217, and 0.234 ± 0.0945; the independent conditional IQP per condition gives 0.194 ± 0.0221, 0.234 ± 0.00774, and 0.483 ± 0.0366; the independent unconditional IQP per condition gives 0.141 ± 0.0364, 0.348 ± 0.0355, and 0.366 ± 0.0227. All spreads are sample standard deviations across the same five seeds. These diagnostics expose errors hidden by Hamming-shell aggregation and are not used to select the reported models.

4. Scaling

The scaling models use the Curie–Weiss encoder on every wire, with no ancillas and open lattices listed in Table V. At size n, the target law is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$. Training uses $\beta \in \{0.60, 0.80, 1.20, 1.40\}$, with 10,000 samples per condition and 40,000 per seed. Evaluation uses $\beta \in \{0.40, 0.95, 1.00, 1.05, 1.60\}$. We use three independent seeds at each size. The original configuration uses cosine Adam with initial learning rate 0.01, final rate 10^{-3}, $b_1 = 0.9$, $b_2 = 0.999$, stabilizer 10^{-8}, and initialization width $s_\theta = 0.1$. Each restart has 5,000 updates, with three restarts per seed. These settings are retained independently of the sixteen-qubit HPO; their transfer or retuning has not been evaluated.

Each plotted MMD estimate averages five independent banks per seed and condition grid, using 2,000 target draws per condition, 512 masks per condition and bandwidth, and 1,024 reference draws per condition in each bank. Banks are reused across iterations, with reference samples shared across bandwidths. The figures show the median and interquartile range across three seed averages. Negative unbiased MMD^2 estimates are retained. Size-dependent bandwidths prevent interpreting absolute MMD^2 values as direct fidelity comparisons across system sizes.

TABLE V. Scaling architectures, including single-qubit and nearest-neighbor pair phases.

Visible qubits	Lattice	Phases
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048